Tutorial 3

Ex. 8.5

Q46

$$\int \frac{1}{(x^{1/3}-1)\sqrt{x}}\,dx$$

$$= 2\int \frac{1}{u^{2/3}-1}\,du$$

$$= 2\int \frac{1}{t^2-1}\cdot 3t^2\,dt$$

$$= 6\int \frac{t^2}{t^2-1}\,dt$$

$$= 6\int 1\,dt$$

$$+ 6\int \frac{1}{t^2-1}\,dt$$

$$= 6t + 6\ln|t-1| - 6\ln|t+1| + C$$

$$x^{1/n} \to u = x^{1/n}$$

$$du = \frac{1}{n}x^{-\frac{(n-1)}{n}}\,dx$$

$$= \frac{1}{n}u^{-(n-1)}\,dx$$

$$u = \sqrt{x}$$

$$du = \frac{1}{2\sqrt{x}}\,dx$$

$$t = u^{1/3}$$

$$dt = \frac{1}{3}u^{-2/3}\,du$$

$$= \frac{1}{3}t^{-2}\,du$$

$$du = 3t^2\,dt$$

$$\frac{t^2}{t^2-1} = \frac{t^2-1+1}{t^2-1}$$

$$= 1 + \frac{1}{t^2-1}$$

$$\frac{1}{t^2-1} = \frac{1}{t-1} - \frac{1}{t+1}$$

Q48

$$\int \frac{1}{x\sqrt{x+9}}\,dx$$

$u = \sqrt{x+9}$

$u^2 = x+9$

$x = u^2 - 9$

$dx = 2u\,du$

$$= \int \frac{1}{(u^2-9)u} \cdot 2u\,du$$

$$= 2\int \frac{1}{u^2-9}\,du$$

$u^2 - 9 = (u-3)(u+3) \Rightarrow$ Partial Frac.

Q50

$$\int \frac{1}{x^6(x^5+4)}\,dx \cdot \frac{x^{\boxed{4}}}{x^{\boxed{4}}}$$

$$= \int \frac{x^4}{x^{10}(x^5+4)}\,dx$$

$$= \frac{1}{5}\int \frac{1}{u^2(u+4)}\,du$$

$u = x^5$

$du = 5x^4\,dx$

Q49

$$\frac{1}{x(x^4+1)} \cdot \frac{x^3}{x^3}$$

$$= \frac{\boxed{x^3}}{x^4(x^4+1)}$$

$u = x^4$

$$\frac{1}{u^2(u+4)} = \frac{A}{u} + \frac{B}{u^2} + \frac{C}{u+4}$$

$$\# \int \frac{1}{x^m(x^n+a)}\,dx$$

If $m+n-1$ is a multiple of n, try $u = x^n$.

Q54

$$\int \frac{\boxed{\sqrt{x}}}{\sqrt{2-\sqrt{x}} + \boxed{\sqrt{x}}}\, dx \to \sqrt{x}\,(\quad \cdots \quad)$$

$$u = \sqrt{2-\sqrt{x}}$$

$$u^2 = 2-\sqrt{x}$$

$$\sqrt{x} = 2-u^2$$

$$x = (2-u^2)^2 \Rightarrow dx = -4u(2-u^2)\,du$$

$$= \int \frac{2-u^2}{u+(2-u^2)} \cdot \left(-4u(2-u^2)\right) du$$

Ex. 8.8

Q32 $\displaystyle\int_0^2 \frac{dx}{\sqrt{|x-1|}} = \int_0^1 \frac{dx}{\sqrt{|x-1|}} + \int_1^2 \frac{dx}{\sqrt{|x-1|}}$

$x=1$

Singular point

$$= \int_0^1 \frac{dx}{\sqrt{-(x-1)}} + \int_1^2 \frac{dx}{\sqrt{x-1}}$$

$$= \lim_{c\to 1^-} \int_0^c \frac{dx}{\sqrt{1-x}} + \lim_{b\to 1^+} \int_b^2 \frac{dx}{\sqrt{x-1}}$$

Q46 $\displaystyle\int_0^1 \frac{dt}{t-\sin t}$

$\square \leqslant \square \quad \square \geqslant \square$

Hint: $\boxed{t \geqslant \sin t} \Rightarrow t - \sin t \geqslant 0$

$0 \leqslant \sin t$ for $0 \leqslant t \leqslant 1$

$-\sin t \leqslant 0$

$t - \sin t \leqslant t$

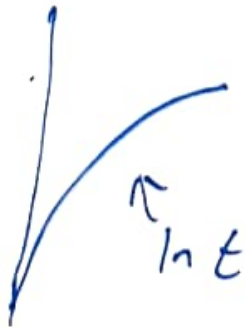
$$\frac{1}{t} \leqslant \frac{1}{t-\sin t}$$

$$\int_0^1 \frac{1}{t}\,dt \leqslant \int_0^1 \frac{1}{t-\sin t}\,dt$$

$$\int_0^1 \frac{1}{t}\,dt = \lim_{c\to 0^+} \int_c^1 \frac{1}{t}\,dt$$

$$= \lim_{c\to 0^+} \Big[\ln|t|\Big]_c^1$$

$$= \boxed{\lim_{c\to 0^+} \ln c = -\infty}$$



Correction!

$$= \lim_{c\to 0^+} (-\ln c)$$

$$= +\infty.$$

$$\int_0^1 \frac{dt}{t-\sin t}$$

$\nearrow f(t), g(t) \geq 0$

$$\lim_{t\to 0^+} \frac{f(t)}{g(t)} = L \quad (0 < L < \infty)$$

Then $\int_0^1 f(t)\,dt$ converges

iff $\int_0^1 g(t)\,dt$ converges.

$t - \sin t \sim \frac{1}{6} t^3$ $\quad$ $\tan x - x \sim \frac{2}{3} x^3$

Set $f(t) = \frac{1}{t-\sin t}$, $g(t) = \frac{1}{t^3}$

$$\lim_{t\to 0^+} \frac{f(t)}{g(t)} = \lim_{t\to 0^+} \frac{t^3}{t-\sin t} \begin{matrix} \to 0 \\ \to 0 \end{matrix}$$

$$= \lim_{t\to 0^+} \frac{3t^2}{1-\cos t} \begin{matrix} \to 0 \\ \to 0 \end{matrix}$$

$$= \lim_{t\to 0^+} \frac{6t}{\sin t} \begin{matrix} \to 0 \\ \to 0 \end{matrix}$$

$$= \lim_{t\to 0^+} \frac{6}{\cos t} = 6$$

$$\int_0^1 g(t)\,dt$$

$$= \lim_{c \to 0^+} \int_c^1 \frac{1}{t^3}\,dt$$

$$= \infty$$

$\therefore \int_0^1 g(t)\,dt$ diverges.

$\Rightarrow \underline{\int_0^1 f(t)\,dt}$ diverges by LCT.

$$\ln \underline{x} \ll \underline{x^p} \ll \underline{\boxed{e^x}}$$

converge

$$\frac{1}{e^x} \ll \frac{1}{x^p} \ll \frac{1}{\ln x}$$

$$\downarrow$$

$$\frac{1}{x^q} \ll \frac{1}{x^p}$$

$q > 1$ $\quad p \leq 1$

<u>Q65</u> $\displaystyle\int_1^\infty \frac{1}{\sqrt{e^x - x}}\,dx$

$$\downarrow$$

$$e^{\frac{x}{2}}$$

$$\lim_{x \to \infty} \frac{f(x)}{g(x)}$$

$$f(x) = \frac{1}{\sqrt{e^x - x}}, \quad \underline{g(x) = \frac{1}{e^{x/2}}}$$